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Biaser, method and system

Abstract

A biaser including a tubular body having a longitudinal axis, an outside surface, and an inside surface. A first plurality of openings extend from the outside surface to the inside surface, with adjacent ones of the openings each defining a beam extending at a first angle other than along the tubular body axis. A second plurality of openings in the body extend from the outside surface to the inside surface, with adjacent ones of the openings each defining a beam extending at a second angle that is other than along the tubular body axis and other than at the first angle. A method for making the biaser, includes depositing material and fusing the material together according to the geometry of the biaser.

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Background/Summary

BACKGROUND

(1) In the resource recovery and fluid sequestration industries biasing devices such as springs are often needed to actuate tools in the downhole environment. Some of these are quite adept at managing the duty to which they are put while others may not be. Unequivocally there are myriad operations for which a biasing device might be helpful but for which no suitable device presently exists. Maximization in efficiencies are always desirable. For this reason, the art is always receptive to advancements.

SUMMARY

(2) An embodiment of a biaser including a tubular body having a longitudinal axis and having an outside surface and an inside surface, a first plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending

at a first angle other than along the tubular body axis, and a second plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending at a second angle that is other than along the tubular body axis and other than at the first angle.
(3) An embodiment of a method for making the biaser, including depositing material in a desired location and fusing the material together according to the geometry of the biaser.
(4) An embodiment of a borehole system including a borehole in a subsurface formation, a string in the borehole, a biaser disposed within or as a part of the string.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
(2) FIG. **1** is a sectional view of a biaser as disclosed herein;
(3) FIG. **2** is a view of a tubular body of the biaser of FIG. **1**;
(4) FIG. **3** is an enlarged section of a portion of the view of FIG. **2**;
(5) FIG. **4** is a view of the tubular body of the biaser in a compressed state;
(6) FIG. **5** is a sectional view of the biaser of FIG. **1** in an as printed condition;
(7) FIG. **6** is the view of FIG. **5** in a completed condition; and
(8) FIG. **7** is a view of a borehole system including the biaser as disclosed herein.
DETAILED DESCRIPTION
(9) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
(10) Referring to FIG. **1**, a sectional view of a biaser **10** is illustrated. Biaser **10** includes a tubular body **12** and may also in some embodiments may optionally include, a second tubular structure **14** that may be positioned either radially inwardly of the body **12** or radially outwardly of the body **12**. As illustrated the structure **14** is positioned radially outwardly of the body **12**. Structure **14** may be configured as a compression limiter, an extension limiter or both as desired in various embodiments. Structure **14** may be integral with one end of the body **12** and mobile with respect to the opposite end of the body **12** or may be geometrically interlocked with an interference fit to one end of the body or may be geometrically loosely interlocked both ends of the body **12**. As used herein, the term “geometrically loosely interlocked” is meant to convey that two components are not separable in the ordinary course from one another but that they can move relative to one another due to clearance in the connection therebetween. For example, a dovetail connection may be considered geometrically loosely interlocked if the dovetail dimensions are loose rather than an interference fit. A loose dimension connection will create a joint that can move about to some degree while still not ordinarily coming completely apart. Structure **14** may be solid walled or may have openings therein as desired. Dimensions of the structure **14** will take into account its construction (integral or not), potential movement afforded in its connection with the body **12**, and its overall longitudinal axial length so that compression of the body **12** is limited to a selected degree and/or extension of the body **12** is limited to a selected degree.
(11) Returning to the body **12**, it will be appreciated from the Figures herein that the body **12** includes a plurality of openings **16** that extend from an outside surface **18** of the body **12** through the body **12** to an inside surface **20** of the body **12**. Between adjacent openings **16** are beams **22**. The openings **16**, and accordingly the beams **22**, are arranged at an angle to a longitudinal axis **24** of the body **12**. Angles range from about 30 degrees to about 75 degrees, and in some cases could range from about 40 to about 60 degrees, from an orthogonal plane of the axis **24** and therefore each define a segment of a helix. In an embodiment, the angle is 45 degrees. In another

embodiment, the angle is 53 degrees. The beams **22** allow for deflection when the body **12** is compressed (see FIG. **4**, for example) or deflection in opposite direction when the body is extended. In either case, the beams are maintained in an elastic range of deformation so that spring rate and repeatability are retained.
(12) Where only one bank **26** of beams **22** exists in the body **12**, a torsional reaction will occur pursuant to axial input whether compressional or tensile. In embodiments, such torsional movement is to be avoided, and in such embodiments, more than one bank **26** of beams **22** is used. In the figures hereof, four banks **26** of beams **22** are illustrated. Even numbers of banks **26** with opposing angles will negate torsional reaction of the body **12**. This example is a simple case and it is to be appreciated that more complex cases can also result in a net zero torsional reaction even if adjacent banks **26** are not mirrored angles. Rather, the net zero torsional reaction merely requires that when all banks **26** in a particular embodiment are summed, the result is zero net torsional reaction.
(13) As illustrated, the banks **26** are separated by a ring **28** of material that is orthogonally positioned relative to the axis **24**. Also, in embodiments, chamfers and/or rounded edges **30** are provided on the beams **22** to reduce stress risers and improve service life of the biaser **10**.
(14) The biaser **10** is manufacturable either substantively or more easily additively. It will be appreciated by those of skill in the art that the angle range of 30 to 75 degrees from the orthogonal to the axis **24** is within the tolerance for a build plate additive manufacturing method and easily can hold tight tolerances. With additive manufacture methods, the structure **14** may be printed along with the body **12**.
(15) Referring to FIGS. **5** and **6**, an embodiment where the structure **14** is radially inwardly disposed of the body **12** is printed to be integral with the body **12**. If left in the condition of FIG. **5**, the biaser **10** would not work because there is a solid connection from one end to the other end through structure **14** that does not provide for deflection. This embodiment is easily post treated however with a simply milling operation to remove material shown at numeral **32** in FIG. **5** and missing in FIG. **6** to that the biaser **10** may be extended and compressed to the degree that material was removed in the milling operation. It should also be appreciated that the illustrated embodiment includes a tensile stopper **34**. Embodiments may or may not include this stopper depending upon whether or not extension limit is desired.
(16) While FIGS. **5** and **6** illustrate only the structure **14** radially inwardly of the body **12** it will be easily appreciated by those of skill that the structure **14** could be radially outwardly disposed and the milling operation noted could occur from the outside of the biaser **10**.
(17) Referring to FIG. **7**, a borehole system **40** is illustrated. The system **40** comprises a borehole **42** in a subsurface formation **44**. A string **46** is disposed within the borehole **40**. A biaser **10** as disclosed herein is disposed within or as a part of the string **46**.
(18) Set forth below are some embodiments of the foregoing disclosure:
(19) Embodiment 1: A biaser including a tubular body having a longitudinal axis and having an outside surface and an inside surface, a first plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending at a first angle other than along the tubular body axis, and a second plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending at a second angle that is other than along the tubular body axis and other than at the first angle.
(20) Embodiment 2: The biaser as in any prior embodiment, wherein the first plurality of openings and the second plurality of openings are longitudinally spaced from one another.
(21) Embodiment 3: The biaser as in any prior embodiment, further comprising additional pluralities of openings longitudinally spaced from one another, each additional plurality of openings having a beam extending in the first or second direction opposite of the next adjacent longitudinally spaced plurality of openings.

(22) Embodiment 4: The biaser as in any prior embodiment, wherein the number of pluralities is even such that the body when compressed or stretched is substantially nontorsional.
(23) Embodiment 5: The biaser as in any prior embodiment, at least one of the plurality of openings is helically arranged.
(24) Embodiment 6: The biaser as in any prior embodiment, wherein the first angle and the second angle are mirror images of one another.
(25) Embodiment 7: The biaser as in any prior embodiment, wherein the first angle is measured from an orthogonal plane to the longitudinal axis of the body is from about 30* to about 75*.
(26) Embodiment 8: The biaser as in any prior embodiment, wherein the first angle and second angle measured from an orthogonal plane to the longitudinal axis of the body is from about 40* to about 60*.
(27) Embodiment 9: The biaser as in any prior embodiment, wherein the second angle measured from an orthogonal plane to the longitudinal axis of the body is from about 30* to about 75*.
(28) Embodiment 10: The biaser as in any prior embodiment, further comprising a second tubular structure disposed adjacent the body and secured to the body to limit compression and/or extension of the body.
(29) Embodiment 11: The biaser as in any prior embodiment, wherein the second tubular structure is separate from the body and geometrically loosely interlocked with the body.
(30) Embodiment 12: The biaser as in any prior embodiment, wherein the second tubular structure is a part of the body at one end of the second tubular structure.
(31) Embodiment 13: The biaser as in any prior embodiment, wherein an opposite end of the second tubular structure is geometrically loosely interlocked with the body.
(32) Embodiment 14: The biaser as in any prior embodiment, wherein the second tubular structure is radially outwardly disposed of the body.
(33) Embodiment 15: The biaser as in any prior embodiment, wherein the second tubular structure is radially inwardly disposed of the body.
(34) Embodiment 16: The biaser as in any prior embodiment, further comprising configuring the plurality of openings to reduce stress risers in the beam.
(35) Embodiment 17: The biaser as in any prior embodiment, wherein the configuring is a rounded end of the beam.
(36) Embodiment 18: The biaser as in any prior embodiment, wherein the configuring is a chamfered end of the beam.
(37) Embodiment 19: A method for making the biaser as in any prior embodiment, including depositing material in a desired location and fusing the material together according to the geometry of the biaser.
(38) Embodiment 20: A borehole system including a borehole in a subsurface formation, a string in the borehole, a biaser as in any prior embodiment disposed within or as a part of the string.
(39) The use of the terms "a" and "an" and "the" and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Further, it should be noted that the terms "first," "second," and the like herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms "about", "substantially" and "generally" are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, "about" and/or "substantially" and/or "generally" includes a range of ±8% of a given value.
(40) The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof.

Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.

(41) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

Claims

1. A biaser comprising: a tubular body having a longitudinal axis and having an outside surface and an inside surface; a first bank having a first longitudinal end of the first bank and a second longitudinal end of the first bank and defining a first plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending at a first angle other than along the tubular body axis and other than orthogonal to the tubular body axis, the first end of the first bank configured to exhibit torsional movement relative to the second end of the first bank upon compression or tension applied to the first bank; and a second bank having a first longitudinal end of the second bank and a second longitudinal end of the second bank and defining a second plurality of openings in the body extending from the outside surface to the inside surface, adjacent ones of the openings each defining a beam extending at a second angle that is other than along the tubular body axis, other than at the first angle, and other than orthogonal to the tubular body axis, the first end of the second bank configured to exhibit torsional movement relative to the second end of the second bank upon compression or tension applied to the second bank.

2. The biaser as claimed in claim 1, wherein the first plurality of openings and the second plurality of openings are longitudinally spaced from one another.

3. The biaser as claimed in claim 1, further comprising additional pluralities of openings longitudinally spaced from one another, each additional plurality of openings having a beam extending in the first or second direction opposite of the next adjacent longitudinally spaced plurality of openings.

4. The biaser as claimed in claim 1, wherein individual banks collectively substantially cancel the torsional movement within the biaser.

5. The biaser as claimed in claim 1, at least one of the plurality of openings is helically arranged.

6. The biaser as claimed in claim 1, wherein the first angle and the second angle are mirror images of one another.

7. The biaser as claimed in claim 1, wherein the first angle is measured from an orthogonal plane to the longitudinal axis of the body is from about 30° to about 75°.

8. The biaser as claimed in claim 1, wherein the first angle and second angle measured from an orthogonal plane to the longitudinal axis of the body is from about 40° to about 60°.

9. The biaser as claimed in claim 7, wherein the second angle measured from an orthogonal plane to the longitudinal axis of the body is from about 30° to about 75°.

10. The biaser as claimed in claim 1, further comprising a second tubular structure disposed adjacent the body and secured to the body to limit compression and/or extension of the body.
11. The biaser as claimed in claim 10, wherein the second tubular structure is separate from the body and geometrically loosely interlocked with the body.
12. The biaser as claimed in claim 10, wherein the second tubular structure is a part of the body at one end of the second tubular structure.
13. The biaser as claimed in claim 12, wherein an opposite end of the second tubular structure is geometrically loosely interlocked with the body.
14. The biaser as claimed in claim 10, wherein the second tubular structure is radially outwardly disposed of the body.
15. The biaser as claimed in claim 10, wherein the second tubular structure is radially inwardly disposed of the body.
16. The biaser as claimed in claim 1, further comprising configuring the plurality of openings to reduce stress risers in the beam.
17. The biaser as claimed in claim 16, wherein the configuring is a rounded end of the beam.
18. The biaser as claimed in claim 16, wherein the configuring is a chamfered end of the beam.
19. A method for making the biaser as claimed in claim 1, comprising: depositing material in a desired location and fusing the material together according to the geometry of the biaser.
20. A borehole system comprising: a borehole in a subsurface formation; a string in the borehole; a biaser as claimed in claim 1 disposed within or as a part of the string.